MILITARY SPECIFICATION

CLEANER, STEAM AND HIGH PRESSURE HOT AND COLD WATER JET, TRAILER MOUNTED

This specification is approved for use by all Departments and Agencies of the Department of the Army.

1 SCOPE

1.1 Scope. This specification covers a steam cleaner, electric motor driven, diesel fuel fired with steam, hot water and cold water cleaning capabilities.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
O-A-548	- Antifreeze/Coolant, Engine, Ethylene Glycol, Inhibited, Concentrated
TT-P-636	- Primer Coating, Alkyd, Wood and Ferrous Metal
MMM-A-260	- Adhesive, Water-Resistant, (for Sealing Waterproofed Paper)
PPP-B-636	- Box, Shipping, Fiberboard
PPP-B-1055	- Barrier Material, Waterproofed, Flexible
PPP-C-843	- Cushioning, Material, Cellulosic
PPP-C-850	- Cushioning Material, Polystyrene, Expanded, Resilient (for Packaging Uses)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Engineering Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using a self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PPP-C-1797 - Cushioning Material, Resilient, Unicellular, Polyethylene Foam
PPP-T-60 - Tape, Packaging, Waterproof

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronics and Associated Equipment)
MIL-C-5501 - Caps and Plugs, Protective, Dust and Moisture Seal, General Specification for
MIL-E-16298 - Electric Machines Having Rotating Parts and Associated Repair Parts: Packaging of
MIL-B-22191 - Barrier Material, Transparent, Flexible, Heat Sealable
MIL-C-52950 - Crates, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-171 - Finishing of Metal and Wood Surfaces
MIL-STD-633 - Mobile Electric Power, Engine Generator Standard Family, General Specifications
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods
MS52152 - Chassis, Trailer; 3/4 Ton, 2 Wheel, Military Design, M116A2

2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ARMY

11024000 - Cleaner, Steam, Press Jet, Hot and Cold Rinse
11024120 - Valve, Solenoid, Water
11024156-2 - Valve, Back Pressure, 400 psi
11024156-4 - Valve, Back Pressure, 600 psi
11024289-2 - Switch, Pressure, 400 psi
11024289-1 - Switch, Pressure, 600 psi
11024780 - Trailer Assembly
11024645 - Cleaner, Steam, High Pressure Jet, Hot and Cold Rinse, Trailer Mounted

(Copies of specifications, standards, handbooks, drawings, and publications required by contractors in connection with specific acquisition functions should be

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obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D2156	- Smoke Density in Fuel Gases from Burning Distillate Fuels
ASTM D3951	- Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 Description. The cleaner shall be in accordance with Drawings 11024645, 11024000, and 11024780 and all subassembly drawings referenced therein and as specified herein.

3.1.1 Drawings. The drawings referenced herein are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the supplier shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the shop equipment. Any data (e.g, shop drawings, layouts, flow sheets, processing procedures, etc) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting office or his designated representative.

3.1.2 Weight. The maximum wet weight (including accessories and fluids) shall not exceed 3,100 pounds (see 4.5).

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 6.2.1, and 6.3).

3.3 Performance.

3.3.1 Steam capacity. With 50 feet of cleaning hose and the cleaning gun attached, burning No. 2 fuel, the cleaner operating on both 50 and 60 Hertz (Hz), single phase, 208 volts (± 10 percent), shall discharge saturated steam at no less than:

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200 gallons per hour (gph)
315 degrees Fahrenheit (F)
200 pounds per square inch gage (psi)

From a cold start with 40 degrees F or colder water, the steam cleaner shall produce the rated output of saturated steam in no more than 5 minutes (see 4.7.4.3.1 and 4.7.4.3.2).

3.3.2 Hot water rinse mode. In the hot water rinse mode, operating on both 50 and 60 Hz, single phase, 208 volts (± 10 percent), using 50 feet of cleaning hose with the gun and the high pressure rinse nozzle attached, the cleaner shall be capable of discharging no less than 240 gph of 180 to 200 degrees F water at no less than 400 psi pressure as measured at the discharge manifold (see 4.7.4.2).

3.3.3 Cold water rinse mode. While in the cold water rinse mode, operating on both 50 and 60 Hz , single phase, 208 volts (± 10 percent), using 50 feet of cleaning hose with the gun and the high pressure rinse nozzle attached, the cleaner shall be capable of discharging no less than 240 gph of inlet water at no less than 400 psi pressure as measured at the discharge manifold (see 4.7.4.1).

3.3.4 Thermal efficiency. In both the hot water and steam modes, the cleaner shall have a thermal efficiency (see 6.7) no less than 75 percent while operating at full capacity. The efficiency shall be met using Number 2 diesel fuel and on both 50 and 60 hertz power sources. The thermal efficiency after 200 hours of operation shall be no less than 70 percent. Smoke density measured in accordance with ASTM D2156 shall be no greater than Number 2 during the first 400 hours of operation. Readjustment of fuel delivery rate is not permitted to satisfy other performance requirements (see 4.7.5.3.1).

3.3.5 Altitude requirements. The cleaner shall be capable of performing as specified herein at altitudes from sea level to 5000 feet. From 5000 feet to 6000 feet the cleaner shall be capable of performing as specified herein with the exception that the requirements shall be met at only 60 Hz and a temperature of +72±5 degrees F. (see 4.7.6.2).

3.3.6 Environmental requirements. The cleaner shall be capable of performing to full capacity as stated herein at temperatures from + 35 degrees F, to + 125 degrees F and relative humidities at 97 to 100 percent. Full performance shall also be obtained after being subjected to storage temperatures of from minus 50 degrees F, for up to 72 hours to plus 150 degrees F for up to 24 hours. The cleaner and power supply mounted on the trailer with all equipment stored in the proper place and with panels and doors closed, shall withstand the simulated rain test in 4.7.6.8, without damage to the cleaner. The cleaner shall be capable of meeting all the operating requirements herein, before and after the simulated rain test (see 4.7.6.3 through 4.7.6.8).

3.3.7 Transportability. The design of the steam cleaner shall incorporate the capabilities to accomodate without degradation (i.e., cracked welds, permanent deformation, damaged equipment, etc.) (see 4.7.5.2.3) the following:

(a) Truck towing at speeds up through 50 miles per hour (mph) on primary roads up through 35 mph on secondary gravel roads and up through 5 mph on unimproved (cross country (see 6.7.3)) terrain by all vehicles from the 5/4 ton CUCV through the 5 ton cargo truck.
(b) The lifting eye shall be capable of lifting 2.5 times the wet weight of the cleaner (see 4.7.6.9).

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3.4 Tilted operation. The cleaner shall show no evidence of liquid spillage or leakage when filled to full liquid levels and shall perform within the capacity requirements specified herein without any disassembly or adjustment when tilted in any of the four major directions 15 degrees from a horizontal plane (see 4.7.5.3.2).

3.5 Shutdown.

3.5.1 Steam mode (Blowdown). The cleaner shall operate automatically through a blowdown (see 6.7.2) cycle every time the cleaner is switched from the steam mode of operation to the OFF position. The coil drain valve shall open automatically. The burner shall continue to operate, and the pump shall continue to operate in the bypass mode for a period of 127±10 seconds. The flow shall then automatically be redirected through the coil and out the drain valve for a period of 88±10 seconds. At that time the pump shall automatically stop and the other drains shall open. (see 4.7.5.3.4).

3.5.2 Hot water and cold water mode. The cleaner shall be operated in both the hot water and cold water modes. For both the hot water and cold water modes when the control switch is turned to the OFF position, the power to the cleaner circuits shall be automatically maintainted for 6±2 seconds and all three drains shall automatically open and drain (see 4.7.5.3.4).

3.6 Construction. The cleaner and its components shall be free from any characteristics or defects which may render the cleaner unsuitable or inefficient for the purpose intended. The cleaner and its equipment shall be so constructed as to be capable of performing the operations specified herein.

3.6.1 Materials. Materials shall be as specified herein and as shown on drawing 11024645 and the applicable drawings referenced therein. Materials not specifically designated shall be selected by the contractor and shall be subject to all provisions of this specification. The contractor is encouraged to use reclaimed materials for fabricating new parts without jeopardizing the chemical and physical properties, design integrity, and intent of the materials originally selected or specified. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner that shall restore them to the same chemical composition and physical properties as the materials originally selected for use. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment.

3.6.2 Threads. All threaded parts of the cleaner and its components shall conform to FED-STD-H28 and applicable "Detailed Standard" section referenced therein.

3.6.3 Fungus and moisture resistance. The electrical circuitry specified on the applicable drawings referenced therein shall be protected from fungus growth and moisture by an overall fungus treatment in accordance with MIL-V-173 with the exception that:

(a) Components of circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely effected by the coating shall not be treated.

3.7 Reliability. The cleaner shall have a minimum acceptable value (MAV) of 150 hours Mean Time Between Failure (MTBF) and a best operation capability (BOC) of 225 hours MTBF.

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3.8 Safety devices. Safety devices shall be as specified herein and on drawing 11024645 and drawings referenced therein.

3.8.1 Power loss. A manually actuated master switch shall provide on/off control for all electrical power to the cleaner. The cleaner shall not restart automatically after a power interruption (see 4.7.3.1).

3.8.2 Burner operation. Fuel flow shall be either prevented or terminated under the following conditions (see 4.7.3.2 through 4.7.3.5):

(a) When there is no flame in the burner combustion chamber.

(1) A flame is not established within 45 seconds during starting (see 4.7.3.2).
(2) A flameout occurs (see 4.7.3.2).

(b) When the water flow from the pump is blocked or reduced to less than 60 gph, except during blowdown cycle, (see 4.7.3.3).

(c) When the fluid temperature at the coil outlet exceeds 220 degrees F in the hot water mode and 340 degrees F in the steam mode. Water shall continue to flow through the coil (see 4.7.3.4).

(d) When the fluid pressure to the coil outlet exceeds 400 psi in the steam mode and 600 psi in the hot water mode (see 4.7.3.5).

3.8.3 Bypass system. When operated in the bypass mode for a period of 30 minutes, the water temperature in the bypass loop shall not exceed 190 degrees F. (see 4.7.3.6).

3.8.4 Safety device settings. When changing from the steam mode to high pressure rinse mode, the high temperature and high pressure limiting devices shall be automatically changed to the correct limiting value (see 4.7.3.4 and 4.7.3.5).

3.8.5 High voltage. The cleaner shall withstand, without damage, at least 1,500 volts rms at 60 Hz, applied to every part of the 208 volt circuit in the cleaner. The voltage shall be maintained for at least 15 seconds between mutually insulated circuits, and between ground and insulated circuits (see 4.7.2).

3.9 Water supply system. The cleaners shall operate from a pressurized water source not exceeding 125 psi and also by drawing unpressurized water from a pond, lake, river, well, etc. The water supply system shall prevent back-siphonage into the water source.

3.9.1 Pressurized water source. Steam cleaners shall take the water from a pressurized water source of from 25 to 125 psi. Each cleaner shall be furnished with 50 feet of hose for connection to a pressurized water source.

3.9.2 Suction supply source. The cleaner shall be capable of drawing water from a lake, stream, well, etc., with a suction head of no less than 15 feet below the water pump inlet (priming is permitted to initiate water flow). Each cleaner shall be equipped with no less than 50 feet of suction hose. (see 4.7.6.1).

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3.10 Smoke density. The burner shall be capable of burning diesel fuel, grades with No. 1 or No. 2, while not emitting a smoke output greater than No. 2 on the Bacharach scale. (see 4.7.5.3.3).

3.11 Cleaning solution. The cleaning solution metering system shall deliver one part of cleaning solution to twelve parts of discharge water at full scale of the metering valve (see 4.7.1).

3.12 Government-furnished property. Unless otherwise specified (see 6.2.1), the following property in the quantities indicated will be furnished by the Government:

Item No.	Description	Identification	Quantity for each cleaner
1	Chassis, Trailer; 3/4 Ton, 2 Wheel, Military Design, M116A2	MS52152	1 each
2	Generator Set, MEP-016-B, 3KW, 60 Hz, Diesel Engine Driven, with Built-In Noise Suppression Feature	MIL-STD-633	1 each

3.13 Modification to Government furnished property trailer. The Government furnished trailer (see 3.12) shall be modified as specified on drawing 11024780.

3.14 Protective finish. Surfaces to be painted shall be cleaned and dried to insure they are free of oil, grease, welding slag and spatter, loose mill scale, water, dirt, corrosion products or any other interfering substances. Cleaning methods or solutions used shall provide a thoroughly clean surface and shall not cause any damage to accompanying or attached parts such as rubber, plastic, leather, or any other natural or synthetic materials. After cleaning and before any corrosion product or other coating interfering material can be deposited, the surface shall be prepared or treated in accordance with MIL-STD-171 to insure adhesion of the coating system. Finish methods shall be as specified on the individual drawing. The dry film thickness shall be as specified in MIL-STD-171. The primer and paint shall be in accordance with the individual drawing and MIL-STD-171.

3.15 Workmanship. Workmanship shall be equal to that in evidence within the cleaner industry. External surfaces shall be smooth and all edges shall be rounded or beveled. Standards of workmanship shall be such that equipment specified herein shall comply with section 3 requirements. All components, subassemblies, parts, and the final assembly shall be capable of performance specified herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

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4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that the products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall be appled to the preproduction model or initial production item noted in 3.2. First article inspection shall consist of the examination in 4.5 and all tests under 4.7. The tests shall be performed in accordance with the test conditions in 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (i) as listed. Failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (4.5).
(b) High voltage requirement test (4.7.2).
(c) Safety controls tests (4.7.3).
(d) Cold water discharge test (4.7.4.1).
(e) Hot water discharge test (4.7.4.2).
(f) Steam discharge test (4.7.4.3).
(g) Reliability test (4.7.5).
(h) Protective finnish test (4.7.7).
(i) Packaging inspection (4.8).

4.4.1 Sampling. Sampling for quality conformance inspection shall be in accordance with MIL-STD-105 using Inspection Levels and Acceptable Quality Levels (AQL) as listed, unless 100 percent inspection is specified. A sample lot shall consist of a quantity produced in one month or 80 cleaners, whichever is less.

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Inspection or Test	Inspection Level	AQL
Product examination (see 4.5)	100 percent	-
High voltage requirement test (see 4.7.2)	S-4	2.5
Safety controls test (see 4.7.3)	100 percent	-
Cold water discharge test (see 4.7.4.1)	100 percent	-
Hot water discharge test (see 4.7.4.2)	100 percent	-
Steam discharge test (see 4.7.4.3)	100 percent	-
Reliability test (see 4.7.5)	*	-
Protective finnish test (see 4.7.7)	S-4	1.0
Packaging inspection (see 4.8)	S-4	4.0

*The reliability test shall be conducted on the first cleaner produced if a first article test is not required by the contract. If a first article test is required, the reliability test shall be performed as part of the first article test.

4.5 Product examination. Visually, manually, and dimensionally examine the cleaner to determine conformance with the requirements of 3.4 through 3.4.3 and 3.10 through 3.13. Visual examinations shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from the identified defects. Manual examinations shall include the operation of moveable parts by hand to assure proper functioning. Dimensional examinations shall include hose lengths and shall include overall dimensions (see 3.1.) including weights and other dimensions as specified. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the contractor to provide objective evidence that the cleaner and its components have passed the examinations prescribed for them by the contractor's inspection system shall be cause for rejection.

4.6 Test conditions. The cleaner shall be tested and inspected under the listed conditions:

4.6.1 Preparation for test. The cleaner shall be prepared for test by servicing, lubricating, adjusting, and running-in as specified by the operator's manual. All controls, valves, and pump(s) shall be checked to assure proper operation. For all discharge tests use 50 feet of discharge hose.

4.6.2 Measurements.

4.6.2.1 Pressure. Pressure shall be measured at the inlet and outlet sides of the heating coil. Pressure shall be measured using calibrated equipment capable of providing accuracy within 1 percent of the measured value, and reading in pounds per square inch gage.

4.6.2.2 Temperature. Temperature measurements shall be taken at the inlet and outlet sides of the heating coil. Temperature shall be measured using calibrated equipment capable of providing accuracy within 1 percent of the measured value, and reading in degrees F. The existing thermocouple on the outlet side of the coil may be used providing it is properly installed per the applicable drawing. A thermocouple may be taped to the coil inlet line only.

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4.6.2.3 Fuel consumption. Fuel usage shall be determined by the weight loss method, and the period between weighings of the fuel shall represent at least five minutes of burner operation. Fuel weight shall be measured on a calibrated scale or balance having an accuracy within 2 percent of the measured value, and reading in pounds.

4.6.2.4 Liquid flow. Liquid flow rates shall be determined by use of a flow meter which shall be of the totalizing and recording type and accurate within 3 percent of the measured value. Readings shall be in gallons per time period. Instantaneous flowmeter readings shall not be used.

4.6.3 Fuel. The cleaner shall be supplied with No.2 diesel fuel for all tests specified herein.

4.6.4 Water supply. The water supply for the tests shall be sufficient to supply the cleaner with a flow of at least 240 gallons per hour. Water under pressure of up to 125 psi may be used for all tests except the suction supply water source test (see 4.7.6.1). The temperature of the water supplied to the cleaner shall be as specified for the individual tests.

4.6.5 Chemical additives. Chemical additives such as cleaning solution or descaling solution shall be used in the cleaner only when specified for the individual tests.

4.7 Tests. Cleaners shall be tested and inspected under the test conditions specified in 4.6 and all subparagraphs thereof.

4.7.1 Cleaning solution metering valve test. Set the metering valve at full open, 12 on the scale. Operate the cleaner in the hot water mode and determine the time required to use one gallon of cleaning solution. During this same time period determine the total gun output in gallons. Failure of the cleaner to discharge cleaning solution at the rate of no less than 18.4 gph at a setting of 12 on the scale shall be cause for rejection. (see 3.11).

4.7.2 High voltage requirement test. A test voltage of at least 1500 volts rms, 60 Hz, shall be applied instantaneously between each and every 208 volt circuit in the unit. The voltage shall be supplied by a transformer having a volt-ampere capacity of at least 500. The voltage shall be applied to an uninsulated point in each circuit and between each circuit and frame or chassis ground. The voltage shall be maintained for at least 15 seconds. When test voltage is applied to the power input terminals the master switch and other involved switches shall be in the closed position. When testing internal circuits, solid state devices may be bridged or removed to prevent damage. Any damage to any part other than solid state components evidenced by malfunction, breakdown, arcing, or other surface or air disharge or arcing in relays, shall be cause for rejection (see 3.8.5). CAUTION: Due to high voltages, care is required while conducting this test.

4.7.3 Safety controls tests. The cleaner safety devices shall be tested as specified. All safety controls tests shall be conducted with the cleaner operating under 60 Hz power. Failure of any of the safety devices to meet requirements shall be cause for rejection (see 3.8).

4.7.3.1 Power loss test. With the cleaner operating in the cold water mode, the cleaner electrical power shall be momentarily interrupted by removing the power cord from its socket and then reconnecting it. The cleaner electrical power shall again be interrupted in the same manner while the cleaner is operating in the hot water mode

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and in the steam discharge mode. Automatic restart of any function of the cleaner after a power interruption in any mode (without resetting the master switch) shall be cause for rejection (see 3.8.1).

4.7.3.2 Burner operation test. The cleaner shall be subjected to the following condition while set up for operation in the hot water mode, and in the steam discharge mode: The continuous spark ignition device shall be displaced from the fuel flow so that the device will continue to operate without causing ignition. An attempt shall then be made to start and operate the cleaner in the normal manner. Failure of the cleaner to automatically stop all flow of fuel to the burner within 45 seconds of the occurance or failure to the burner to re-start without activating the manual reset on the burner assembly shall be cause for rejection (see 3.8.2). The ignition device shall then be returned to its normal position and the cleaner started.

4.7.3.3 Inadequate water flow test. Inadequate water flow test shall be conducted in the steam and hot water modes. The cleaner overtemperature and overpressure safety devices shall be disconnected or otherwise defeated. The water supply to the cleaner shall gradually be reduced. Failure of the cleaner to automatically stop all flow of fuel to the burner before the water flow rate falls below 60 gph or failure of the burner to automatically restart when the water supply is restored shall be cause for rejection. Following this test, all cleaner safety devices shall be returned to their original state (see 3.8.2).

4.7.3.4 Temperature limiting devices test. With the cleaner operating in the hot water mode, the device(s) controlling the water temperature shall be defeated or adjusted so as the water temperature increases, fuel flow to the burner shall automatically stop before water temperature reaches 225 degrees F. If the fuel flow is automatically terminated, the cleaner shall be allowed to cool down until the burner automatically re-starts. The cleaner shall then be operated in the steam mode, and the device(s) controlling the steam temperature defeated or adjusted so as the steam temperature increases,fuel flow to the burner shall automatically stop before the water temperature reaches 350 degrees F. If the fuel flow is automatically terminated, the cleaner shall be allowed to cool down until the burner automatically re-starts. Failure of the cleaner to automatically stop all flow of fuel to the burner before the water temperature exceeds 225 degrees F in the hot water mode and before the steam temperature exceeds 350 degrees F in the steam mode; failure of the cleaner to maintain the flow of water through the coil after fuel flow has been terminated; or failure of the burner to automatically re-start after the temperature falls back within acceptable limits shall be cause for rejection (see 3.8.2 and 3.8.4). NOTE: If necessary, the cleaner pressure regulating device(s) may be bypassed or taken out of the circuit for the duration of this test. At the conclusion of this test all safety devices shall be returned to their original state.

4.7.3.5 Pressure limiting devices test. Attach a pump to the cleaning hose outlet connection that is capable of delivering a back pressure of at least 4 gpm flow at pressure up to 650 psi. Conduct tests as listed with special instructions as follows:

(a) For Test 1 and 3, use an appropriate continuity tester across the electrical contacts to determine when the valves open.
(b) For Test 3, energize Solenoid, PN 11024120 in coil coutlet lines with 110 volts.

Test No.	Test Part Number	Apply Press.	Apply Flow	Opening Pressure Range - psi
1	11024289-2	X	-	375 - 425

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2	11024156-2	X	4gpm	375 - 425
3	11024289-1	X	-	575 - 625
4	11024156-4	X	4gpm	575 - 625

Return all parts to original state. Start the cleaner and operate in the hot water mode. Disconnect one of the wires to the pressure switch listed in Test 1. Fuel flow to the burner shall stop within 45 seconds and start flow when the wire is reconnected. In the steam mode, disconnect one of the wires to the pressure switch listed in Test 3. Fuel flow to the burner shall stop within 45 seconds and start when the wire is reconnected. Failure of any of the safety devices to meet the requirements shall be cause for rejection (see 3.8.2 and 3.8.4).

4.7.3.6 Pump bypass test. After the hot water discharge test (4.7.4.2) conduct the pump bypass test. With the cleaner in operation in the hot water mode, stop discharge flow by shutting off the gun for a period of no less than 30 minutes. Temperature in the bypass loop in excess of 190 degrees F shall be cause for rejection (see 3.8.3).

4.7.4 Performance tests.

4.7.4.1 Cold water discharge test. The cleaner shall be operated in the cold water mode for at least 15 minutes under 60 Hz power and for at least 15 minutes under 50 Hz power. For the first article test only, the temperature of the inlet water shall be less than 50 degrees F for at least 5 minutes of both the 60 Hz and 50 Hz portions of the test. Failure of the cleaner to continuously discharge at least 240 gph of water through the rinse gun at a pressure of at least 400 psi for the full period of operation under either source of power shall be cause for rejection (see 3.3.3).

4.7.4.2 Hot water discharge test. The cleaner shall be operated in the hot water mode under 60 Hz power for 15 minutes after the water temperature at the coil outlet has stabilized and for the same period under 50 Hz power. During the first article test only, the temperature of the inlet water shall be 40 degrees F or less for the duration of both the 60 Hz and 50 Hz portions of the test. Failure of the cleaner to continuously discharge at least 240 gph of water heated to a temperature of at least 180 degrees F at a pressure of at least 400 psi for 15 minutes while operating under either source of power shall be cause for rejection (see 3.3.2).

4.7.4.3 Steam discharge tests.

4.7.4.3.1 Cold start test. The cleaner shall be configured for operation at 50 Hz power. Operate in the cold water mode for 5 minutes to insure that the coil is cool. For the first article test only, the cleaner shall then be set for operation in the steam mode and supplied with water at a temperature of not more than 40 degrees F. Turn on and operate for five minutes. Failure of the cleaner to discharge steam heated to a temperature of at least 315 degrees F at a pressure of at least 200 psi and a rate of at least 200 gph by the end of the five minute period shall be cause for rejection (see 3.3.1). On first article test, check thermal efficiency at this point. Thermal efficiency of less 75 percent shall be cause for rejection (see 4.7.5.3.1).

4.7.4.3.2 Operational test. The cleaner shall be operated in the steam mode under 50 Hz power. The cleaner shall be operated for at least 12 minutes after the temperature at the coil outlet has stabilized. During this period of time pressures at the pump outlet and at the pressure gun shall be monitored and recorded at 3 minute intervals. Failure of the cleaner to continuously discharge steam heated to a temperature of at least 315 degrees F at a pressure of at least 200 psi and at a rate

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of at least 200 gph for the full 12 minute period shall be cause for rejection (see 3.3.1).

4.7.5 Reliability test (see 3.5).

4.7.5.1 Test conditions and data. The cleaner shall be tested and data taken under the listed conditions.

4.7.5.1.1 Preparation for test. The cleaner shall be trailer mounted. Ambient air temperature shall be between 35 and 115 degrees F and the water source shall be capable of providing 240 gallons per hour (gph) at a temperature between 35 and 120 degrees F.

4.7.5.1.1.1 Shop environment test. The cleaner shall be operated by a 208 VAC plus or minus 10 percent, 50 Hz power source. Inlet water pressure shall be between 35 and 125 psi.

4.7.5.1.1.2 Field environment test. The cleaner shall be powered by the self contained generator at 208 plus or minus 10 percent at 60 Hz. Water shall be pulled from a non-pressurized tank or a stream. Inlet water temperature shall be between 35 and 120 degrees F.

4.7.5.1.1.3 Transport test. For road test all test equipment shall be removed to avoid damage. All regular equipment shall be stowed in the proper place and all doors shut.

4.7.5.1.2 Data readings. Provide data sheets with provision for recording cleaner serial number and date of test. The following readings shall be taken at intervals of not more than one hour:

(a) Time of day.
(b) Cleaner hour meter reading.
(c) Fluid pressure at inlet and outlet of heating coil using equipment with an accuracy of 1 percent of the measured value in psi.
(d) Fluid temperature at inlet and outlet of heating coil using equipment with an accuracy of 1 percent of the measured value and reading in degrees F.
(e) Liquid flow rates determined by using flow meters which shall be of the totalizing and recording type and accurate within 3 percent of the measured value and shall read in gallons per time period for water and pounds per time period for fuel. Instantaneous meter readings shall not be used.
(f) Fuel flow rate as determined by weight loss method over a period of five minutes. Calibrated scale shall have an accuracy within 2 percent of measured value and reading in pounds with the flow rate to be in pounds per minute.
(g) Input voltage using equipment having an accuracy of 1 percent of measured value.
(h) Amperage using equipment having an accuracy of 2 percent of measured value.
(i) Frequency in Hz using equipment having a accuracy of 2 percent of measured value.

4.7.5.2 Reliability test missions. Reliability test on the cleaner shall be conducted in mission increments consisting of modes and times listed for a minimum of 473 hours and a maximum of 1020 hours (see 4.7.5.4.). During the test record data at intervals of not more than one hour as specified in 4.7.5.1.2. Modes are:

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4.7.5.2.1 Shop environment mode. Operate for a total of twelve hours per mission. Operate 5.0 hours in the steam mode, 1.0 hour of which shall be with chemical solution with the metering valve set at 1. Operate 6.0 hours in the hot water mode, 4.8 hours of which shall be with chemical solution with the metering valve set at 1. Operate 1.0 hour in the cold water mode. All operations shall be performed under the conditions listed in 4.7.5.1.1.1.

4.7.5.2.2 Field environment mode. Operate for a total of eight hours per mission. Operate 3.0 hours in the steam mode, 0.6 hour of which shall be with chemical solution with the metering valve set at 1. Operate 4.0 hours in the hot water mode, 3.2 hours of which shall be with chemical solution with the metering valve set at 1. Operate 1.0 hours in the cold water mode. All operations shall be performed under conditions listed in 4.7.5.1.1.2.

4.7.5.2.3 Road tow test mode. Tow cleaner using appropriate vehicle, over 6.3 miles of primary (paved) road at a safe speed not to exceed 50 mph, over 1.1 miles of secondary (gravel) road at a safe speed not to exceed 35 mph, and 0.6 mile of unimproved (cross country (see 6.7.3)) at a safe speed not to exceed 5 mph (see 3.3.7).

4.7.5.3 Tests during reliability test. During the reliability testing conduct other tests as follows:

4.7.5.3.1 Thermal efficiency. At the end of each 200 plus or minus 8 hours of the test, after the cleaner has operated for at least 10 minutes in the steam mode, measure the following parameters, then calculate the thermal efficiency to verify it is no less than 70 percent, using the following:

Calculation - Thermal efficiency:

$$E = \frac{(Qw)\ (T)\ (C)\ (D)\ (100)}{(Qf)\ (H)}$$

Where:

Qw = water flowing through coil (gal).
C = specific heat of water (1.0 BTU/lbm F).
D = density of water (8.33 lbm/gal).
Qf = fuel consumed (lbm).
H = energy content of fuel (BTU/lbm).
T = difference between coil outlet and inlet water temperatures (F).

4.7.5.3.2 Tilted operation test. During the initial mission testing and while all tanks are full, conduct angle operation tests by tilting the trailer with the cleaner 15 degrees in one of the four major directions. Operate at this angle for the full 8 hours in the steam mode during the shop environment mode. Repeat with the trailer tilted in the other three directions on subsequent missions. Verify proper cleaner operation at all four angles and there shall be no spillage from any tank while tilted (see 3.4.7).

4.7.5.3.3 Smoke density test. Conduct a smoke density test within 10 hours of the start of the reliability test and at every 300±10 hours of the reliability test. Verify the density is no greater than a No. 2 smoke spot when measured in accordance with ASTM D2156 (see 3.10).

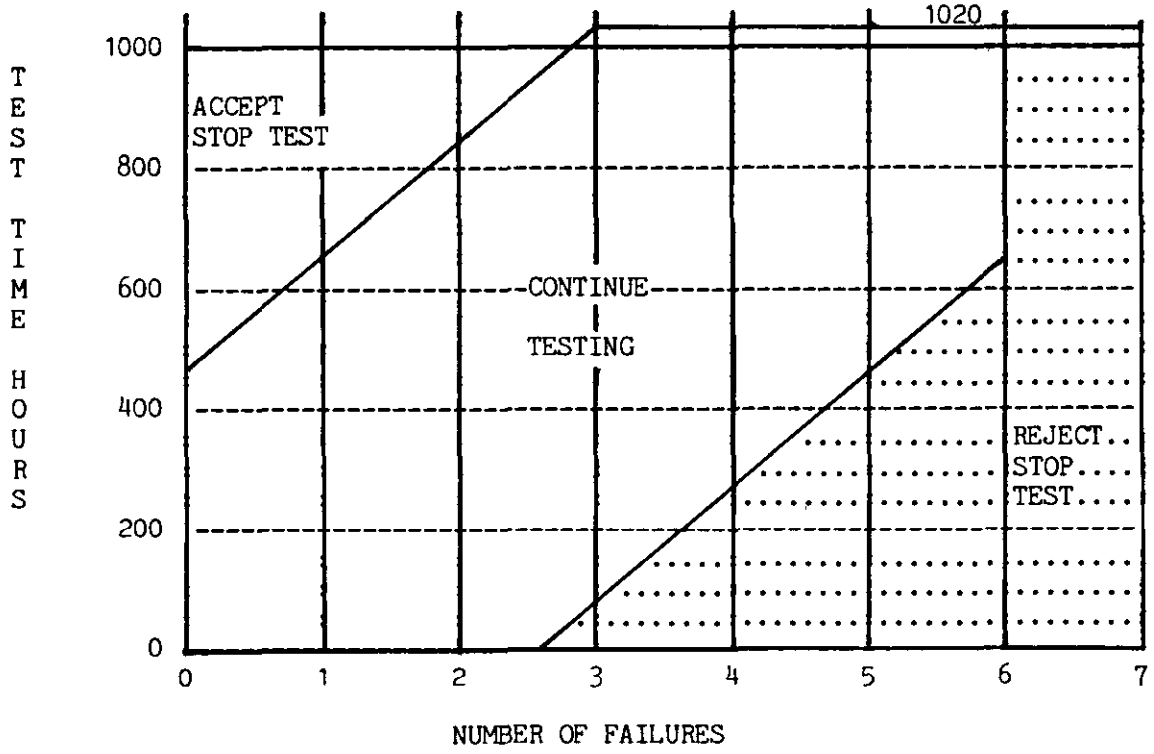
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4.7.5.3.4 Shutdown tests. With the cleaner operating in the steam mode, turn the control switch to the OFF position. Verify the coil drain valve opens, the burner continues to operate, and the pump operates in the bypass mode. This will continue for 127±10 seconds. The flow in then redirected thru the coil and out the drain valve for a period of 88±10 seconds. At that time the pump will stop, the other drains will open, and all cleaner functions will stop. Verify all three drains are open and draining. With the cleaner operating in both the hot water and cold water mode, turn the control switch to the OFF position. Verify that power to the cleaner circuits is maintainted for 6±2 seconds and that all three drains are open and draining (see 3.5).

4.7.5.4 Reliability test duration. The cleaner shall be operated on a mission basis with each mission being 20 hours of operation plus transportation, which does not add to total hours. Operate for a minimum of 473 hours with no failures or for a maximum of 1020 hours with a maximum of six failures for acceptance. For definition of a failure see 4.7.5.5. The following table shall be used to determine the future of the test:

Test Time Hours	Number of Failures: Accept, Stop Test	Continue Testing	Reject, Stop Test
73	-	0 to 2	3
255	-	0 to 3	4
437	-	0 to 4	5
473	0	1 to 5	6
655	1	2 to 5	6
837	2	3 to 5	6
1020	6	-	7

The following is a chart of the table for clarification purposes. When failures fall in the shaded area the test shall be terminated.



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4.7.5.5 Definition of a failure. A failure is generally defined as any physical or mechanical malfunction which the operator cannot remedy by adjustment, repair or replacement action using available parts and tools within thirty minutes and which causes or may cause: (1) Inability to commence operation, (2) Cessation of operation, (3) Degradation of operator/equipment capability below designated levels, (4) Serious personnel hazards. Malfunctions which do not affect mission preformance will not be considered mission failures (see 3.5).

4.7.5.6 Maintenance information.

(a) Date, time, and cleaner hourmeter reading at which the action was initiated.
(b) Type of maintenance, servicing, or adjustments performed and manhours to accomplish.
(c) Date, time, and cleaner hourmeter reading when the action was completed.

4.7.5.7 Corrective action. Failure of a cleaner to meet the continuous duty test requirement shall be cause for deferring further submission of cleaners for Government acceptance until the contractor has completed the following actions:

(a) Conduct a failure analysis study, performing a dimensional, physical, and visual examination, as required, of the component(s) or condition(s) suspected to be the cause of failure.
(b) Evaluate and correct the applicable production processes and procedures to prevent recurrence of the same defect in future production.
(c) Examine cleaners awaiting submittal for Government acceptance, parially assembled cleaners, and components (including components and subassemblies at in-process or final assembly) to insure that material containing the same defect(s) is purged from the inventory and not presented to the Government for acceptance.
(d) Provide objective evidence of the results of the failure analysis and corrective actions to the Government representative prior to submitting the reconditioned test sets for retest.

4.7.6 Environmental and function tests.

4.7.6.1 Suction supply water source. With the 50 foot suction hose, with the foot valve attached, the cleaner shall be operated under 50 Hz power in the cold water mode to draw water against a resistance equivalent to a 15 foot vertical rise (priming is permitted to initiate water flow). This may be accomplished either by drawing from a standing body of water whose surface is at least 15 feet lower in elevation than the water pump inlet, or by drawing against an equivalent resistance of 15 feet of head at standard atmospheric pressure at sea level. Failure of the cleaner to draw the water and discharge it at a rate of at least 240 gph and a pressure of at least 400 psi shall be cause for rejection (see 3.9.2).

4.7.6.2 Altitude test. The cleaner shall be placed in a test chamber or tested under conditions to maintain a pressure equivalent to a 6,000-foot altitude and a temperature of 72±5 degrees F. It shall then be subjected to the tests in 4.7.4.2 and 4.7.4.3.2 with the exception that the test shall be performed at 60 HZ only. Scientific computation based on stoichiometric calculations, fuel analysis and other test parameters shall be approved by the procuring activity prior to this test if submitted in lieu of conducting this test. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2 with the exception that the test shall be performed at 60 HZ

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only under the above conditions shall be cause for rejection (see 3.3.5).

4.7.6.3 High operating temperature test. The cleaner, after being in an ambient air temperature of 120 degrees F to 125 Degrees F until stabilized, and while at that temperature, shall be tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2 while in the above condition shall be cause for rejection (see 3.3.6).

4.7.6.4 Low operating temperature test. The cleaner, after being in an ambient air temperature of 35 degrees F to 40 degrees F until stabilized, and while at that temperature, shall be tested in accordnace with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2 while in the above condition shall be cause for rejection (see 3.3.6).

4.7.6.5 High storage temperature test. The cleaner and acessories shall be submitted to a 24 hour cycle of temperature starting with ten hours at 80 plus or minus 5 degrees F, followed by five hours of steady increase to 150 plus or minus 5 degrees F, four hours at temperature, and five hours of steady decrease to 80 plus or minus 5 degrees F. The equipment shall then be inspected for evidence of damage and tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2, or evidence of damage to any cleaner component after exposure to the above conditions shall be cause for rejection (see 3.3.6).

4.7.6.6 Low storage temperature test. The cleaner and accessories shall be placed in an ambient air temperature of -50 plus or minus 5 degrees F for a period of 72 hours. At the end of this time, the equipment shall be removed and allowed to stabilize at an air temperature of 80 plus or minus 5 degrees F. The equipment shall then be inspected for evidence of damage and tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2, or evidence of damage to any cleaner component after exposure to the above conditions shall be cause for rejection (see 3.3.6).

4.7.6.7 Humidity test. The cleaner and accessories shall be subjected to a 24 hour exposure of 97 to 100 percent relative humidity at 80 degrees F to 85 degrees F air temperature, air movement and radiation effects negligible. The cleaner shall then be tested in accordance with 4.7.4.2 and 4.7.4.3.2. Failure of the cleaner to pass the tests of 4.7.4.2 and 4.7.4.3.2, or evidence of damage to any cleaner component after exposure to the above conditions shall be cause for rejection (see 3.3.6).

4.7.6.8 Simulated rain test. The cleaner and power supply shall be mounted on the trailer with all equipment stowed in the proper place and with panels and doors closed. Provide a lawn sprinkler that will apply simulated rain which produces droplets of uniform distribution and velocity similar to natural rain at the point of contact with the cleaner falling toward the cleaner at angle of approximately 45 degrees to the surfaces. The volume of rain impinging on the surface shall be not less than two inches in a five minute period. Direct the water for five minutes on each of the four sides of the cleaner assembly. Turn off the water and prepare cleaner for operation by removing panels. Operate in cold water, hot water and steam modes for at least 5 minutes each. Failure of the cleaner to operate properly in any of the three modes shall be cause for rejection (see 3.3.6).

4.7.6.9 Lifting tests. Sufficient weight, uniformly distributed, shall be added to the trailer to bring the total weight to 2.5 times the wet weight (the weight of the cleaner, power unit, and trailer with all equipment and all tanks full of fluid). The unit shall then be slowly lifted by the lifting eye approximately one foot and

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suspended for at least 30 seconds. There shall be no damage to the lifting eye or any other parts of the unit. Damage shall be cause for rejection (see 3.3.7).

4.7.7 Protective finish test. The hardness, adhesion and thickness of the paint used on the machine shall be determined (see 3.14), as follows:

(a) The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the unit until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.
(b) A test sample of the same metal, painted with the same process and at the same time as the unit, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure of the surface to pit (paint too hard) shall be cause for rejection.
(c) Two, two-inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of 1/2 inch wide fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.
(d) Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least the thickness specified on the applicable drawings shall be cause for rejection.

4.8 Packaging inspection. The items shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing and marking requirements specified in Section 5.

5 PACKAGING

5.1 Disassembly. Cleaners shall be disassembled to the extent necessary to assure that all surfaces requiring a preservative are accessible for processing, to assure minimum cube for packing, and to permit reassembly without special skills or tools. Disassembly shall be in accordance with MIL-STD-1186.

5.2 Preservation. Preservation shall be military level A or industrial, as specified in the contract (see 6.2.1).

5.2.1 Military level A.

5.2.1.1 Cleaning and drying. The cleaner components and accessories, and all tanks, tubing, piping, fittings, and valves shall be drained, cleaned, dried, and treated with preservative prior to submission for acceptance. All procedures shall be in accordance with the applicable procedures of MIL-P-116.

5.2.1.2 Preservation application. Immediately after cleaning and dryng, apply a coat of type P-2 preservative of MIL-P-116 to all exterior, critical, bare, ferrous metal surfaces of the cleaner. All surfaces coated with P-2 shall be covered with barrier material conforming to MIL-B-121 and secured with tape conforming to PPP-T-60. Exterior, noncritical, bare, ferrous metal surfaces, which have not been painted or plated, shall be coated with type P-1 or type P-19 preservative. Interior surfaces of the delivery gun(s) shall be preserved with type P-10, grade 30, preservative oil. Interior surfaces of the fuel system shall be completely flushed (lines and internal filters) and completely fogged (tank and filter(s)) with type P-10, grade 30, preservative oil. Interior surfaces of the water system shall be completely flushed with a solution consisting of 50 percent antifreeze (ethylene glycol) conforming to O-A-548 and 50 percent clean fresh water. Induce sufficient air pressure into the

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system to remove residual water.

5.2.1.3 Unit protection. The basic cleaner shall be unit packaged in accordance with method I of MIL-P-116. Small openings in the cleaner shall be sealed with tape or plastic cap-plugs conforming to PPP-T-60 or MIL-C-5501, respectively. Large apertures shall be covered with barrier material conforming to PPP-B-1055 (class E-2). Secure barrier material in place with above mentioned tape. Components of the cleaner and accessories shall be provided unit protection as specified in paragraphs 5.2.1.3.1 through 5.2.1.3.8 below.

5.2.1.3.1 Gages and meters. Gages and meters shall be covered with 1 inch thick uncompressed cushioning material conforming to PPP-C-1797, type I or PPP-C-850, type I or II, class 1 or 2, grade R. Secure cushioning in place with PPP-T-60 tape. When gages and meters are protected by an integral cover, cushioning material will not be required.

5.2.1.3.2 Drive belts. Drive belts shall be either removed or released from tension. The grooves of all ferrous metal pulleys shall be coated with primer conforming to TT-P-636. Belts removed from the cleaner shall be packaged in accordance with method III of MIL-P-116.

5.2.1.3.3 Electric motors. Electric motors shall be packaged in accordance with level A requirements of MIL-E-16298 and the applicable requirements for assembled machines.

5.2.1.3.4 Switch boxes. Electric switch boxes shall be completely sealed with PPP-T-60 tape.

5.2.1.3.5 Power cable and delivery hoses. The ends of the hoses and power cable shall be wrapped with barrier material conforming to MIL-B-22191, type III. Secure material with PPP-T-60 tape by encircling the hose or cable, as applicable, making a water-tight closure. Individually, coil the hose and cable to a minimum uniform, compact, safe diameter to prevent any deformation or kinking. Secure coils with a minimum of three ties of cotton tape or twine to prevent uncoiling. Alternatively, the hoses and cable may be coiled on their respective stowage racks and secured to prevent loosening.

5.2.1.3.6 Tools and delivery nozzles. Maintenance tools and delivery nozzles shall be wrapped with barrier material conforming to MIL-B-22191, type III or L-P-378, type III, secured with tape, and placed in their respective stowage compartments.

5.2.1.3.7 Cleaning gun. Cleaning guns shall be placed and secured in the stowage device provided on the cleaner. Apertures shall be sealed with tape conforming to PPP-T-60, type IV.

5.2.1.3.8 Publications. Technical publications issued with the cleaners shall be packaged in accordance with method IC-1 of MIL-P-116.

5.2.1.3.9 Consolidated packaging. All packaged accessories, technical publications, and disassembled parts, which have not been placed in stowage racks or applicable compartments of the cleaner, shall be adequately cushioned with material conforming to PPP-C-843, type II and packaged in one or more snug-fittings fiberboard boxes conforming to PPP-B-636, class weather resistant, type CF. The container shall be waterproof sealed, as specified in the appendix to the box specification.

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5.2.2 Industrial preservation. Industrial preservation shall be as specified in ASTM D3951.

5.3 Packing. Packing shall be military level A or industrial, as specified in the contract (see 6.2.1).

5.3.1 Military level A. Each cleaner shall be packed in an open crate conforming to MIL-C-52950, type V, style A. The consolidated package shall be placed inside the shipping container and all items secured to restrict any movement. Arrangements of contents, anchoring, blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. The cleaner shall be shrouded with barrier material conforming to PPP-B-1055, class E-2, H-5, or M-1. Adhesive used in the shroud fabrication shall conform to MMM-A-260. Application of the shroud shall be in accordance with the appendix to the crate specification. Polyvinyl chloride (PVC) shall not be acceptable as shroud material because of the possible corrosive effect of PVC on the shrouded item.

5.3.2 Industrial. Industrial packing shall be in accordance with ASTM-D-3951.

5.4 Marking. Marking shall be military or industrial as specified in the contract (see 6.2.1).

5.4.1 Military. Marking of interior packages and exterior shipping containers shall be as specified in MIL-STD-129 and any special marking, as required therein. Shipping container shall also be marked with "DO NOT STACK" in at least two places (both ends of both sides). Bar code marking is required and shall be in accordance with MIL-STD-129.

5.4.2 Industrial. Industrial marking of interior packages and exterior shipping containers shall be as specified in ASTM-D-3951. Shipping containers shall also be marked with "DO NOT STACK" in at least two places (both ends or both sides). Bar code marking is required and shall be in accordance with MIL-STD-129.

6 NOTES

6.1 Intended use. The steam cleaner covered by this specification is intended for cleaning vehicle components prior to maintenance or repair.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article, when required (see 3.1).
(c) Government furnished properity required (see 3.10).
(d) Selection of applicable level of preservation and packing and type of marking (see 5.2, 5.3, and 5.4).

6.3 First article. When a first article inspection is required, the cleaner(s) should be a first article sample, a first production item, a sample selected from the first run of production items or it may be a standard production item from the contractor's current inventory as specified in 4.3. The first article should consist of a specific number of units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles.

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Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering products which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Contract data requirements. Required technical data such as operator's manuals, parts lists, and other instructions for operation and maintenance, as identified on a numbered DD Form 1664, should be specified on DD Form 1423 incorporated in the contract.

6.5 Repair part manuals. Repair part identification, in repair part manuals, shall include the prime contractor's name, nomenclature, part number, and price. Repair part identification shall also include the name of the original equipment/part manufacturers (CAGE), nomenclature, part number, and price of the part immediately after the prime contractor's part identification. High mortality rate items shall be so identified as shown in MIL-M-7298.

6.6 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emmission, noxious vapors, heat, etc., as applicable, specific requirements concerning such points of operation and other health and safety requirements, should be specified by the user.

6.7 Definitions.

6.7.1 Thermal efficiency. Thermal efficiency is defined as the ratio of the heat energy units added to the discharged water divided by the heat energy units released by the oxidation of the fuel.

6.7.2 Blowdown. Blowdown is defined as the removal of solids from the heating coil by the use of a combination of heat and water pressure through an automatic cycle that activates timers, thermocouples, relays and valve openings.

6.7.3 Cross country. Cross country consists of native soil with rock added as required to maintain passability. Surface conditions range from smooth to rough with wash-boarding, rutting, potholes and depressed areas.

6.8 Subject term (key word) listing.

Cleaner, steam
Cleaning
Construction
Generator
Pump
Safety
Workmanship

Custodian:
Army - AL

User Activities
Army - ME
Navy - YD

Preparing Activity
Army - AL

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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PREVIOUS EDITION IS OBSOLETE.

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